Outline of the First Four Weeks

- **Topic:** Generative AI on Text data (LLMs)
  - Fundamentals of Natural Language Processing (NLP) and Large Language Models (LLMs)
  - Text generation and its evaluation
  - Applications: RAG, PEFT, Reasoning, Agentic AI, Hallucination, …
- **Assignment** (Due: 4/10)
  - You can choose one among the following topics:
    - **Fine-tuning** your own model using PEFT
    - Customized **RAG** for your downstream task
    - Using **LLM Agents** for Collaborative Solutions to Complex Problems

Policy for Assignment (1)

- There are NO restrictions on the use of **generative AI tools** to assist with tasks such as planning strategies or coding during the assignment.
- However, the scope and purpose of any generative AI usage must be clearly reported in the **final report**.
- Directly using topics, content, or repositories posted on the internet without proper adaptation may result in serious penalties.
- You must choose a topic and submit a **proposal report** outlining which model, which dataset, and which task you will use for the assignment. (Due: 3/17)
  - Ex) Dialogue system that provides emotional support for individuals in need of mental assistance

    Domain-specific research assistant for technical papers

    Multi-step decision support system for planning problems

Policy for Assignment (2)

- The project must include either the incorporation of external knowledge and modifications appropriate to the task/data to the baseline method.
  - Ex) What additional capabilities should be given to existing models to better support people in need of mental assistance?

    The document format is irregular. What techniques should I use to parse it effectively?

  - A modification of appropriate difficulty is required.
    - **Bad example:** I told LLM to cheer up and its performance improved. I can't analyze the result any more..

    - **Good example:** Based on prior research [Doe et al., 2012], I hypothesized that specific psychological factors influence the speakers in this task. I therefore compared generation performance across multiple keyword conditions reflecting these factors and injected this information into the model using methods such as prompt prefixes and cross-attention. The results showed that generation quality was highest under [---] condition, and a qualitative study of the outputs confirmed that the characteristics of the utterances changed accordingly. These findings suggest that …

Policy for Assignment (3)

- What the **final report** must include:
  - **Performance & Ablation Study:** Thoroughly report the impact of your modifications. Compare quantitative and qualitative results against the baseline method.
  - **Qualitative Analysis:** Use actual generated outputs to analyze the effectiveness and limitations of your approach.
- A significant performance improvement may be difficult to achieve and is not strictly required.
- Even if the quantitative performance is not strong, it is important to document the efforts you made to improve performance and analyze why the changes worked or did not work.